IBM DA Professional Certificate
Course 5: Python Project for Data Science

Module 1: Data Extraction via APIs and Web Scraping

This module establishes the architectural framework for automated data acquisition. Success in the final certification requires mastery of the request-response cycle, the internal mechanics of the `yfinance` library, the logic of Document Object Model (DOM) traversal, and the mitigation of risks associated with non-deterministic web environments.

1.1 Programmatic Extraction: The yfinance API Architecture

The `yfinance` library serves as a Pythonic interface for Yahoo Finance's internal REST API. It abstracts the complexity of HTTP headers and session management into discrete objects.

- **The Ticker Object & Metadata Retrieval:**
  - Initializing a `Ticker` instance (e.g., $msft = yf.Ticker("MSFT")$) creates a local handle for a specific equity.
  - **The .info Dictionary:** This attribute performs an asynchronous fetch of a JSON payload containing approximately 150 distinct metadata fields.
  - **Data Integrity:** API-sourced data is statistically superior to scraped data because it follows a strict schema, resulting in a $40–60\%$ reduction in data cleaning latency.
- **Historical Data Extraction (.history()):**
  - **Temporal Parameters:** `period` defines the duration (e.g., $1d, 5d, 1mo, 1y, ytd, max$).
  - **Resolution Parameters:** `interval` defines the frequency of data points (e.g., $1m, 2m, 5m, 1h, 1d, 1wk, 1mo$).
  - **Precision Logic:** The method returns a Pandas DataFrame utilizing 64-bit floating-point precision. For a standard 10-year daily historical query, the memory footprint is approximately $250\ \mathrm{KB}$ to $300\ \mathrm{KB}$.
  - **Auto-Adjustment:** By default, `yfinance` adjusts price data for splits and dividends to maintain time-series continuity.

1.2 Web Scraping Mechanics: Structural Parsing of HTML

When a governed API is unavailable, data must be harvested from the presentation layer (HTML) using the `requests` and `BeautifulSoup` libraries.

- **The HTTP Request Protocol:**
  - **GET Requests:** Utilizing $requests.get(url)$ initiates a handshake with the server.
  - **Status Code Audit:** * **200:** Success.
    1. **403:** Forbidden (often triggered by missing User-Agent headers).
    2. **404:** Resource Not Found.
    3. **503:** Service Unavailable (possible rate-limiting).
  - **Latency Benchmark:** Raw HTML retrieval is network-bound. Average latency for a commercial financial page is 800ms to 2500ms
    .
- **Document Object Model (DOM) Traversal:**
  - **BeautifulSoup Object:** Converts raw strings into a navigable tree structure.
  - **Parser Selection:** $html.parser$ is standard; $lxml$ is preferred for high-speed processing of large ($> 2\ \text{MB}$) HTML files.
  - **Selection Logic:**
    1. **find():** Locates the first occurrence of a specific tag (e.g., $< h1 >$).
    2. **find_all():** Returns a list-like object (ResultSet) of all occurrences.
    3. **Attributes:** Accessing internal tag data (e.g., $row.find("td")["style"]$).

1.3 Computational Complexity & Memory Management

- **Parsing Overhead:** BeautifulSoup creates a dense object tree. A $1\ \text{MB}$ HTML file can consume
  $10\text{–}20\ \text{MB}$ of RAM once parsed into a BeautifulSoup object due to the metadata associated with each node.
- **Search Complexity:** Searching the DOM tree using $find_all$ has a time complexity of $O(N)$, where N is the total number of tags in the document.
- **Optimization Strategy:** To minimize the search space, analysts should first locate a unique container (e.g., $id = "revenue - table"$) before iterating through child nodes.

1.4 Data Transformation: Semi-Structured to Structured

The extraction pipeline concludes with the sanitization and loading of data into a Pandas DataFrame.

- **Iterative Row Extraction:**
  - Locate the $< table >$ or $< tbody >$ tag.
  - Loop through $< tr >$ (Table Row) tags.
  - Extract text from each $< td >$ (Table Data) tag using the $.text$ attribute.
  - **Sanitization:** Apply $.strip()$ to remove whitespace and "ν" characters.
  - **Type Casting:** Convert string-based financial figures (e.g., $"1,200.50"$) to floats by removing commas and dollar signs.
- **Pandas read_html Engine:**

- $pd.read_html(url)$ is a high-level wrapper that automates the extraction of all $< table >$ elements.
- **Constraint:** This method is non-deterministic on pages with complex JavaScript rendering or nested table structures.

1.5 Ethics and Performance Constraints (The robots.txt Audit)

- **Governance:** The `robots.txt` file (located at $domain.com/robots.txt$) provides the legal directives for automated agents.
- **Crawl-Delay:** A directive specifying the minimum time (in seconds) between requests to prevent server-side denial of service.
- **Extraction Resilience:** Professional scripts incorporate $try - except$ blocks to handle missing tags, preventing a single malformed row from crashing a $1,000$-row extraction process.

Technical References & Academic Audits

Recommended Books

- **"Web Scraping with Python"** by Ryan Mitchell. Essential for mastering advanced DOM navigation and handling dynamic content.
- **"Python for Finance"** by Yves Hilpisch. Focuses on the quantitative extraction and management of financial time-series.
- **"Fluent Python"** by Luciano Ramalho. Chapters on context managers and data model protocols relevant to file and network I/O.

Validated Industry Resources

- **yfinance Repository:** Technical API Specifications. Documentation on the Ticker object and data intervals.
- **BeautifulSoup Documentation:** Searching the Tree. Official logic for the find and find_all methods.
- **W3C Standards:** HTML Living Standard. The architectural foundation for DOM tree structures.
- **Requests Library:** Developer Interface. Benchmarks for timeout handling and status code resolution.

Key Terminology Index

- **API (Application Programming Interface):** A protocol enabling structured data exchange between software components.
- **DOM (Document Object Model):** A hierarchical representation of an HTML document as a tree of objects.

- **JSON (JavaScript Object Notation):** A lightweight, text-based data-interchange format.
- **Schema:** The formal structure of a database or data object.
- **Ticker:** A unique alphabetic identifier for a specific stock (e.g., GOOGL, AAPL).

Module 2: Dashboarding and Visualizing Stock Data

This module focuses on the synthesis of extracted datasets into interactive analytical interfaces. Mastery for the certification requires an understanding of the Plotly graphing framework, the logic of shared temporal axes, and the performance trade-offs of web-based rendering engines.

2.1 The Visualization Stack: Plotly vs. Matplotlib

Financial dashboards require high interactivity for "Point-in-Time" inspection.

- **Plotly Graph Objects (go):** * Utilizes a JSON-based schema to render charts in the browser.
    - **Interactivity Benefit:** Supports native hover-labels, zooming, and range-sliders without re-executing Python code.
    - **Performance Metric:** Plotly can efficiently handle up to $50,000–100,000$ data points before browser-side latency (DOM lag) becomes noticeable.
- **Matplotlib/Seaborn Comparison:** * While Matplotlib is faster for static, high-density academic plots, it lacks the native "Web-Ready" interactivity required for modern financial dashboards.

2.2 The make_graph Utility: Functional Abstraction

In the capstone project, a modular function $make_graph$ is typically used to synchronize disparate datasets (Stock Price vs. Company Revenue).

- **Architectural Logic of make_subplots:**
    - **shared_xaxes=True:** Essential for ensuring that the timeline for stock prices perfectly aligns with the timeline for revenue reports.
    - **vertical_spacing:** Typically set to 0.03 to maximize the vertical "Data-Ink Ratio" on the dashboard.
- **Trace Implementation:**
    - **Scatter Traces:** Used for both line and marker plots.
    - **Name Assignment:** Traces must be explicitly named (e.g., $"SharePrice"$ or $"Revenue"$) to populate the interactive legend.

2.3 Temporal Synchronization & Data Alignment

A critical challenge in financial dashboarding is the "Granularity Mismatch" between datasets.

- **Daily vs. Quarterly Disparity:** * Stock price data is typically **Daily** ($n \approx 252$ points/year).
    - Revenue data is typically **Quarterly** ($n = 4$ points/year).
- **Alignment Logic:**
    - Plotly handles this through its internal X-axis scaling. By providing a shared date-time index, the sparse revenue points are plotted alongside the dense price points.
    - **Mathematical Constraint:** To ensure accurate visualization, dates must be cast to $datetime64[ns]$ format in Pandas. String-based dates will result in "Categorical Axis" errors, breaking the temporal continuity of the trendline.

2.4 Dashboard UI Components

- **Range Sliders:** Provides the ability to "Drill-Down" into specific historical periods (e.g., the 2008 Financial Crisis or the 2020 Market Volatility) without reloading data.
- **Range Selectors:** Pre-defined buttons (e.g., $1m, 6m, YTD, 1y, all$) that utilize standard financial reporting windows.
- **Annotations:** Used to highlight specific events (e.g., Product Launches, CEO changes) that correlate with price fluctuations.

2.5 Performance and Memory Optimization

- **JSON Serialization:** Plotly converts Python objects into JSON strings for browser rendering.
    - *Metric:* For a dashboard with $5,000$ data points, the JSON payload is approximately 150 KB. This is highly efficient for remote web-deployment.
- **Data Downsampling:** If plotting "Max" historical data (e.g., 40 years of daily data), analysts should use Pandas $.resample('W').mean()$ to reduce the point count by 80% while maintaining the primary trend, significantly improving rendering speed.

Technical References & Academic Audits

Recommended Books

- **"Interactive Data Visualization with Python"** by Abha Belorkar. Detailed guide on Plotly and Dash architecture.
- **"Python for Finance"** by Yves Hilpisch. The standard text for time-series visualization and financial data engineering.
- **"Storytelling with Data"** by Cole Nussbaumer Knaflic. Essential for the aesthetic and cognitive optimization of dashboard layouts.

Validated Industry Resources

- **Plotly Documentation:** Subplots in Python. Technical breakdown of shared axes and layout grids.
- **Pandas Documentation:** Time Series / Date Functionality. Critical for temporal data alignment.
- **SEC EDGAR System:** Financial Filing Access. The primary source for raw revenue data for web scraping.
- **Towards Data Science:** Visualizing Stock Trends. Quantitative analysis of charting performance.

Key Terminology Index

- **Subplot:** A smaller plot that is part of a larger visualization grid.
- **Trace:** A single set of data points and their visual representation in Plotly.
- **Shared Axis:** A design where multiple plots use the same scale for the X or Y axis.
- **Resampling:** The process of changing the frequency of time-series observations.
- **Granularity:** The level of detail represented by each individual data point.

Module 3: Capstone Project: Final Analysis and Submission Logic

This module focuses on the end-to-end execution of the stock market analysis dashboard. Passing the certification requires a successful synthesis of extraction, sanitization, and visualization, culminating in a peer-reviewed submission.

3.1 The End-to-End Project Workflow

The final project requires the construction of a four-stage data pipeline:

1. **API Acquisition:** Utilizing `yfinance` to extract price history for high-volatility equities (e.g., Tesla or GameStop).
2. **Structural Scraping:** Extracting quarterly revenue data from HTML-based SEC-style financial filings.
3. **Data Sanitization:** Converting string-based financial figures (e.g., removing `","` and `$`) and enforcing `datetime64` temporal types.
4. **Dashboard Synthesis:** Invoking the `make_graph` function to produce a dual-axis comparative visualization.

3.2 Technical Benchmarks for Success

- **API Consistency:** Historical share price data must be captured for the same period as the available revenue data (typically since the company's IPO).
- **Regex-based Cleaning:** Students often fail the quiz on data cleaning logic. The specific transformation required for revenue involves:

    - Removing comma delimiters: $str.replace(',','')$
    - Removing currency symbols: $str.replace('\$','', regex = True)$
    - Dropping Nulls: $df.dropna(inplace = True)$
- **Plotting Integrity:** The final dashboard must display **Revenue** as the secondary subplot and **Share Price** as the primary.

3.3 Troubleshooting the IBM Cloud Lab Environment

The Skills Network Labs (SN Labs) environment has specific architectural constraints that can lead to script failure.

- **Kernel Crashes:** Often caused by attempting to scrape too many pages simultaneously. Keep the request frequency below 2 requests per second.
- **Package Conflicts:** Ensure that `html5lib` or `lxml` is explicitly installed if the default `html.parser` fails to reconstruct the DOM correctly.
- **Memory Management:** In a browser-based Jupyter environment, large BeautifulSoup objects ($> 5 \text{ MB}$) should be deleted ($del \text{ soup}$) after extraction to free up JVM memory.

3.4 Submission & Peer-Review Rubric

The final grade is based on five critical checkpoints:

1. **Ticker Identification:** Did the student extract the correct equity ticker?
2. **Web Scraping Logic:** Is the revenue table extracted correctly from the HTML?
3. **Cleaning Protocol:** Are the revenue values converted to floats?
4. **Graph Execution:** Does the graph show both share price and revenue on a shared timeline?
5. **Insights:** Does the student identify any visual correlation between revenue milestones and price spikes?

3.5 Analytical Audit: Beyond the Lab

In a professionalcontext, students should recognize the **yfinance** limitations:

- **Latency:** It is not a real-time feed; it is delayed by $15–20$ minutes.
- **Survivorship Bias:** The library primarily covers currently listed companies.

- **Accuracy:** Scraped revenue data can be skewed by non-standard reporting years (Fiscal vs. Calendar). Professional analysts validate scraped results against official SEC 10-K filings.

Technical References & Academic Audits

Recommended Books

- **"Python Data Science Handbook"** by Jake VanderPlas. (Essential for mastering the Pandas/Matplotlib/Plotly integration).
- **"Automate the Boring Stuff with Python"** by Al Sweigart. (Excellent chapters on practical web scraping logic).
- **"Analysis of Financial Time Series"** by Ruey S. Tsay. (Mathematical foundation for the variances shown in the capstone).

Validated Industry Resources

- **Plotly Documentation:** Range Sliders and Selectors. Technical details for interactive timeline controls.
- **Pandas User Guide:** Cleaning and Preparing Data. Primary source for regex string manipulation.
- **BeautifulSoup documentation:** Searching the tree. Logical framework for `find_all` and CSS selectors.
- **IBM Skills Network:** Project Submission Portal. Benchmark for passing scores and rubric details.

Key Terminology Index

- **Resample:** The process of changing the frequency of time-series observations (e.g., daily to weekly).
- **Regex (Regular Expression):** A sequence of characters that forms a search pattern for text manipulation.
- **Subplot:** A secondary coordinate system within a single figure.
- **ETL (Extract, Transform, Load):** The standard three-step process for data engineering.
- **Volatility:** The degree of variation of a trading price series over time.

Module 4: Professional Validation, Peer-Review Loops, and Portfolio Strategy

This module focuses on the transition from a completed lab to a professional-grade analytical asset. Success in the final certification stage involves mastering the statistical validation of

results, navigating the peer-review scoring mechanics, and optimizing the project for career impact.

4.1 Statistical Validation of Financial Insights

A completed dashboard is only useful if the trends observed are statistically grounded.

- Correlation Analysis (r): * Students must evaluate the relationship between Revenue growth and Share Price movement.
    - **Metric:** A correlation coefficient (r) where $|r| > 0.7$ suggests a strong linear relationship.
- Coefficient of Determination (R^2):
    - Analysts use R^2 to determine how much of the variance in the stock price is explained by revenue changes.
    - **Validation:** If $R^2 < 0.4$, revenue is a weak predictor, and other variables (market sentiment, interest rates) are dominant.

4.2 The Peer-Review Calibration: Scoring Mechanics

The IBM certification utilizes a double-blind peer-review system. Understanding the "Median Score" logic is vital for final grade optimization.

- **The Median Logic:** To eliminate "Outlier Bias" (extremely harsh or extremely lenient graders), Coursera often uses the median of three peer reviews to determine the final grade.
- **Rubric Fidelity:** Every chart must have a visible title and axis labels. Failure to include a title results in an automatic -1 point deduction per chart in most rubrics.
- **Constructive Critique:** When grading peers, identifying **Syntax Errors** in their web-scraping logic provides the "Loop-back Learning" necessary for the student's own mastery.

4.3 Technical Debt and Project Scalability

A Jupyter Notebook in a sandbox environment is not a production system. Professionals must account for "Technical Debt."

- **API Limitations:** Free APIs like `yfinance` can be deprecated. A robust project should include a **Mock Data Fallback** (e.g., a local CSV) to ensure the dashboard remains functional during outages.
- **Time Complexity of Extraction:** If scraping N financial metrics, the process is $O(N)$. For a portfolio of 500 stocks, sequential scraping becomes a bottleneck ($500 \times 2\text{s} = 1000\text{s}$). Professionals utilize **Asynchronous I/O** (e.g., `asyncio` or `threading`) to reduce execution time by $> 70\%$.
  .

4.4 Professional Portfolio Strategy: The "STAR+" Method

To "crack" the certification impact on the job market, the project must be converted into a "STAR+" (Situation, Task, Action, Result + Metrics) resume bullet.

- **Quantitative Resume Formatting:**
  - *Standard:* "Analyzed Tesla stock data using Python."
  - *Professional (STAR+):* "Developed an automated ETL pipeline extracting 10 years of financial data for Tesla and GameStop; synthesized an interactive Plotly dashboard that visualized revenue-to-price correlation with a 95% confidence interval."
- **GitHub Integration:** * The notebook should be exported as a $README.md$ or hosted via **GitHub Pages**.
  - Including a $requirements.txt$ file (detailing exact versions of `pandas`, `yfinance`, and `plotly`) is the industry benchmark for **Reproducibility**.

4.5 The Final Certification Exam: High-Yield Concepts

- **Library Dependencies:** Remember that `BeautifulSoup` requires an external parser like `lxml` for optimal performance.
- **Data Structures:** Historical data is always returned as a **Pandas DataFrame** where the `Index` is the `Date`.
- **Graphing Constraints:** `make_subplots` is a function from the `plotly.subplots` module, not the core `plotly.graph_objects`.

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Module 5: Final Review & Advanced Troubleshooting for Certification Success

This module serves as a technical "pre-flight check" to resolve the most common blockers encountered in the final assessment and the subsequent certification exam.

5.1 The "AttributeError" Audit

Students frequently encounter $AttributeError:' NoneType' object has no attribute' find'$ during the scraping phase.

- **Root Cause:** The $requests.get()$ call was blocked or the HTML structure changed, resulting in a null BeautifulSoup search result.
- **Mitigation:** Always validate the response status code and use an $if \text{ element is not None:}$ guard before attempting text extraction.
- **Quantitative Risk:** Approximately 15% of financial pages utilize dynamic $class$ names to deter scrapers; students must target stable tags like $< thead >$ or $< tbody >$.

5.2 Pythonic Optimization: Vectorized Cleaning

Avoid using for loops to clean financial strings in DataFrames. Vectorized Pandas operations are significantly faster.

- **Comparative Efficiency:**
  - Loop-based cleaning for 10^5 rows: $\sim 2.5\text{s}$.
  - Vectorized cleaning ($df['Revenue'].str.replace()$): $\sim 0.04\text{s}$.

- **Benchmark:** Vectorization provides a $60–100\times$ performance increase by delegating operations to the underlying C-level NumPy arrays.

5.3 Shared Axis Scaling Errors

When synchronizing Revenue (millions) with Stock Price (dollars), the visualization can become unreadable if the Y-axes are not independent.

- **Correction Logic:** Use $secondary_y = True$ within the add_trace call.
- **Validation:** The final project is graded on whether the **Share Price** is legible alongside **Revenue**. If the Share Price trend appears as a flat line at the bottom, the student failed to implement the dual-axis scale.

5.4 Environment Persistence & Dependency Tracking

In the Skill Network Lab environment, installed packages (like $yfinance$) do not persist between sessions.

- **Certification Blueprint:** Candidates must be able to identify the correct $pipinstall$ commands for $yfinance, plotly,$ and $beautifulsoup4$.
- **Library Versionality:** Incompatibilities between $pandas$ and $xlrd$ can break the $read_excel$ functionality. For the exam, know that $openpyxl$ is the modern default engine.

5.5 Final Certification Exam: High-Yield Logic

- **Ticker Creation:** $yf.Ticker("ticker_symbol")$.
- **Revenue Table Selection:** Usually index $[0]$ or $[1]$ when using $pd.read_html$.
- **Closing Balance:** The $history()$ method returns a column named $Close$, not $Closing_Price$.
- **Data Type:** Historical stock data returned by `yfinance` has an index of type $DatetimeIndex$.

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- **"Python Data Science Handbook"** by Jake VanderPlas. (Essential for mastering the Pandas/Matplotlib/Plotly integration).
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